

## IONIC CONDUCTION IN ETHYLENE OXIDE-PROPYLENE OXIDE COPOLYMERS CONTAINING $\text{LiClO}_4$

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Ethylene oxide and propylene oxide were copolymerized at various mixing ratios using coordination catalysts. Total ionic conductivities and lithium ion transport numbers of the copolymer- $\text{LiClO}_4$ -complexes were measured in the temperature range  $+25^\circ\text{C}$  to  $+90^\circ\text{C}$ . Total conductivity was found to decrease with increasing PO content of the copolymer, however being higher than the conductivities of commercial PEO and PPO. The lithium transport number does not significantly deviate from that of pure polymer- $\text{LiClO}_4$ -complexes.

### 1. Introduction

Relatively high ionic conductivities have been reported for polymer complexes of alkali metal salts with polyethers such as poly(ethylene oxide) (PEO) and poly(propylene oxide) (PPO) [1]. Due to the possible commercial applications of these materials as solid electrolytes, interest in these polymers has greatly increased. Knowledge about the ion conductive behavior in the polyether complexes is at a very early stage, however, and the conductivity of a given polymer cannot be theoretically predicted. The ionic mobility is believed to be correlated with the segmental mobility of the polymer main chain, which within the same polymer family is related to the free volume and the size of the side chains. Here amorphism was systematically changed by varying the monomer ratio in ethylene oxide-propylene oxide copolymers. Ionic conductivities for various ratios are presented.

### 2. Experimental

#### 2.1. Copolymers

The polymeric organotin butyl phosphate coor-

dination catalyst was synthesized according to Nakata [2] by condensing dibutyltin dichloride and tributyl phosphate (both supplied by Ega-Chemie) at  $220$  to  $240^\circ\text{C}$ . Distilled monomers EO (Fluga AG) and PO (Ega-Chemie) were dried to less than 10 ppm of water content over a molecular sieve. Copolymerization reactions were carried out in sealed glass bottles at  $25^\circ\text{C}$  with stirring. The amount of the pulverized, thermally activated catalyst was 3 wt%. Polymers, which were obtained as solid precipitates or slurries, were dissolved in ethanol, filtrated and dried in vacuum at  $60^\circ\text{C}$  for 8 h. The monomer ratios in the copolymers were calculated from integrated  $^1\text{H}$  spectra in 30% 1,3,5-trichlorobenzene solution at  $60^\circ\text{C}$  using a Jeol JNM GX-400 FT NMR spectrometer (table 1).

The average molecular weights were estimated using gel permeation chromatography (RT Waters-410) with RI detector at  $20^\circ\text{C}$  and Microstyragel  $10^6$ - $10^5$ - $10^4$ - $10^3$  Å column system (Water Associates Inc.). The carrier solvent was THF (1.0 ml/min). Narrow-distribution polystyrenes were used as the elution standards. The molecular weights (under 25000) were confirmed using vapor pressure osmometry (Knauer), THF as solvent and biphenyl as a standard (table 1).

Table 1  
Properties of EO-PO copolymers.

| EO/(EO+PO)<br>ratio | $M_w$<br>(g/mol) | $T_g$<br>(°C) | $T_m$<br>(°C) |
|---------------------|------------------|---------------|---------------|
| 1                   | 22000            | n.a.          | 61            |
| 0.974               | 32000            | -54           | 59            |
| 0.860               | 28000            | -59           | 48            |
| 0.765               | 10000            | -64           | 37            |
| 0.696               | 20000            | -67           | 4             |
| 0.470               | 5000             | -67           | n.a.          |
| 0                   | 53000            | -67           | 47            |

Table 2  
Properties of PEO-PPO-LiClO<sub>4</sub> electrolytes.

| EO/(EO+PO)<br>ratio | $T_g$<br>(°C) | $T_m$<br>(°C) | $\log(\sigma/S\text{ cm}^{-1})$<br>(25°C) | $t_{Li+}$ |
|---------------------|---------------|---------------|-------------------------------------------|-----------|
| 1                   | -24           | 96            | -6.1                                      | 0.3       |
| 0.974               | -23           | 92            | -6.9                                      | 0.3       |
| 0.860               | -15           | 98            | -6.7                                      | n.a.      |
| 0.765               | -23           | 101           | -6.5                                      | 0.2       |
| 0.696               | -18           | 101           | -6.8                                      | n.a.      |
| 0.470               | -7            | 102           | -7.5                                      | n.a.      |
| 0                   | -6            | 89            | -7.1                                      | n.a.      |

## 2.2. Polymer electrolytes

The polymers were vacuum dried at 40°C overnight. Lithium perchlorate, LiClO<sub>4</sub> (Fluka), was melted twice in vacuum and transferred to dry argon atmosphere where it was weighed. Tetrahydrofuran, THF (Merck), was dried using molecular sieves.

Polymer electrolytes were prepared by dissolving the polymer and a calculated amount of salt in THF, which was then evaporated. The oxygen to lithium ratio was chosen to be 9. A polypropylene filter paper (Gelman Sciences) was used as an inert matrix. The mean thickness of the samples was 110 µm.

Thermal behavior of the copolymers and the electrolytes was detected in nitrogen atmosphere by Mettler 3000 DSC. Heating rate was 10°C/min.

The total conductivities of the samples were measured using the ac impedance method. The electrolyte film was placed between two stainless steel electrodes held together at constant pressure. The cell was kept in vacuum. Resistive heating was applied at both electrodes over the temperature range of 25 to 100°C. A constant potential of zero volts was applied through a Toho Potentiostat 2000, which was used in conjunction with a NF S-5720B frequency analyzer. The instrumentation was controlled with a Hewlett-Packard 816 computer.

Lithium ion transport number was measured using a polarization cell [3]. The sample was mounted between two lithium electrodes (lithium foil 100 µm from Lithco) at a standard potential of 50 mV. The current was detected using a Toho Potentiostat 2000. The temperature of the sample during this experiment was 50°C.

## 3. Results and discussion

Table 2 shows the results for the glass transition temperatures, melting points, conductivities of the complexes at 25°C, and lithium ion transport numbers.

Though amorphism of the pure polymers (table 1) was increasing with increasing PO content, the glass transition temperature of the electrolyte (table 2) was found to increase with increasing PO content, which indicates decreasing amorphism. Evidently side chains together with the lithium salt make the copolymer crystallize. The Arrhenius plot of the ionic conductivity for some of the polymers is shown in fig. 1.

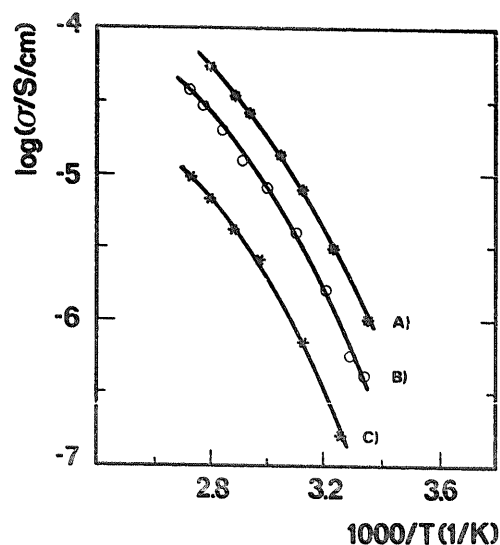


Fig. 1. Relation between logarithm of ionic conductivity and reciprocal of absolute temperature. EO/(EO+PO) ratios are (A) 1, (B), 0.765 and (C) 0.470.

Total conductivity was found to decrease with increasing PO content of the polymer. The decrease was not monotonous as was expected, this may be due to the differences in molecular weights of different copolymers. Total conductivities were slightly higher than those of commercial PEO and PPO. The lithium ion transport numbers were comparable to literature values [4].

## References

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